*Proof.* The Bessel inequality follows from the second stability estimate in (4.7) in combination with the representation in (4.11). The second statement follows from the Pythagoras theorem (4.6) and the representation (4.11). ■

## 4.3 Fourier Partial Sums

In this section, we study one concrete example for Euclidean approximation. In this particular case, we wish to approximate a continuous $2\pi$-periodic function by real-valued trigonometric polynomials. To this end, we equip the linear space of all *real-valued* continuous $2\pi$-periodic functions

$$\mathscr{C}_{2\pi} \equiv \mathscr{C}_{2\pi}^{\mathbb{R}} = \{f : \mathbb{R} \longrightarrow \mathbb{R} \,|\, f \in \mathscr{C}(\mathbb{R}) \text{ and } f(x) = f(x+2\pi) \text{ for all } x \in \mathbb{R}\}$$

with the inner product

$$(f,g)_{\mathbb{R}} = \frac{1}{\pi}\int_0^{2\pi} f(x)g(x)\,\mathrm{d}x \qquad \text{for } f,g \in \mathscr{C}_{2\pi}, \tag{4.13}$$

which by $\|\cdot\|_{\mathbb{R}} = (\cdot,\cdot)_{\mathbb{R}}^{1/2}$ defines a Euclidean norm on $\mathscr{C}_{2\pi}$, so that

$$\|f\|_{\mathbb{R}}^2 = \frac{1}{\pi}\int_0^{2\pi} |f(x)|^2\,\mathrm{d}x \qquad \text{for } f \in \mathscr{C}_{2\pi}.$$

Therefore, $\mathscr{C}_{2\pi}$ with $\|\cdot\|_{\mathbb{R}} = (\cdot,\cdot)_{\mathbb{R}}^{1/2}$ is a Euclidean space.

For the approximation space, we consider choosing the linear space of all real-valued trigonometric polynomials of degree at most $n \in \mathbb{N}_0$,

$$\mathcal{T}_n \equiv \mathcal{T}_n^{\mathbb{R}} = \operatorname{span}\left\{\frac{1}{\sqrt{2}}, \cos(j\,\cdot), \sin(j\,\cdot) \,\middle|\, 1 \le j \le n\right\} \qquad \text{for } n \in \mathbb{N}_0.$$

Therefore, by using the notations introduced at the outset of this chapter, we consider the special case of the Euclidean space $\mathcal{F} = \mathscr{C}_{2\pi}$, equipped with the norm $\|\cdot\|_{\mathbb{R}} = (\cdot,\cdot)_{\mathbb{R}}^{1/2}$, and the linear approximation space $\mathcal{S} = \mathcal{T}_n \subset \mathscr{C}_{2\pi}$ of finite dimension $\dim(\mathcal{T}_n) = 2n+1$, for $n \in \mathbb{N}_0$.

**Remark 4.10.** In the following chapters, we also use *complex-valued* trigonometric polynomials from $\mathcal{T}_n^{\mathbb{C}}$ for the approximation of *complex-valued* continuous $2\pi$-periodic functions from

$$\mathscr{C}_{2\pi}^{\mathbb{C}} = \{f : \mathbb{R} \longrightarrow \mathbb{C} \,|\, f \in \mathscr{C}(\mathbb{R}) \text{ and } f(x) = f(x+2\pi) \text{ for all } x \in \mathbb{R}\}\,.$$

In that case, we equip $\mathscr{C}_{2\pi}^{\mathbb{C}}$ with the inner product

$$(f,g)_{\mathbb{C}} = \frac{1}{2\pi}\int_0^{2\pi} f(x)\overline{g(x)}\,\mathrm{d}x \qquad \text{for } f,g \in \mathscr{C}_{2\pi}^{\mathbb{C}}, \tag{4.14}$$

and thereby obtain the Euclidean norm $\|\cdot\|_{\mathbb{C}} = (\cdot,\cdot)_{\mathbb{C}}^{1/2}$ on $\mathscr{C}_{2\pi}^{\mathbb{C}}$. The different scalar factors, $1/\pi$ for $(\cdot,\cdot)_{\mathbb{R}}$ in (4.13) and $1/(2\pi)$ for $(\cdot,\cdot)_{\mathbb{C}}$ in (4.14), will be useful later. To keep notations simple, we will from now use $(\cdot,\cdot) = (\cdot,\cdot)_{\mathbb{R}}$ and $\|\cdot\| = \|\cdot\|_{\mathbb{R}}$ for the inner product (4.13) and the norm on $\mathscr{C}_{2\pi} \equiv \mathscr{C}_{2\pi}^{\mathbb{R}}$. □

For the approximation of $f \in \mathscr{C}_{2\pi}$, we use fundamental results, as developed in the previous section. In particular, we make use of *orthonormal systems* to construct best approximations to $f$. To this end, we take note of the following important result.

**Theorem 4.11.** *For $n \in \mathbb{N}_0$, the real-valued trigonometric polynomials*

$$\left\{ \frac{1}{\sqrt{2}}, \cos(j\,\cdot), \sin(j\,\cdot) \,\middle|\, 1 \leq j \leq n \right\} \tag{4.15}$$

*form an orthonormal system in $\mathscr{C}_{2\pi}$.*

*Proof.* From the usual addition theorems for trigonometric polynomials we get the identities

$$2\cos(jx)\cos(kx) = \cos((j-k)x) + \cos((j+k)x) \tag{4.16}$$
$$2\sin(jx)\sin(kx) = \cos((j-k)x) - \cos((j+k)x) \tag{4.17}$$
$$2\sin(jx)\cos(kx) = \sin((j-k)x) + \sin((j+k)x). \tag{4.18}$$

The $2\pi$-periodicity of $\cos((j \pm k)x)$ and $\sin((j \pm k)x)$ implies

$$(\cos(j\,\cdot), \cos(k\,\cdot)) = \frac{1}{2\pi}\int_0^{2\pi} [\cos((j-k)x) + \cos((j+k)x)]\,\mathrm{d}x = 0$$
$$(\sin(j\,\cdot), \sin(k\,\cdot)) = \frac{1}{2\pi}\int_0^{2\pi} [\cos((j-k)x) - \cos((j+k)x)]\,\mathrm{d}x = 0$$

for $j \neq k$ and

$$(\sin(j\,\cdot), \cos(k\,\cdot)) = \frac{1}{2\pi}\int_0^{2\pi} [\sin((j-k)x) + \sin((j+k)x)]\,\mathrm{d}x = 0$$

for all $j, k \in \{1, \ldots, n\}$. Moreover, we have

$$\left(\frac{1}{\sqrt{2}}, \cos(j\cdot)\right) = \frac{1}{\sqrt{2}\pi}\int_0^{2\pi} \cos(jx)\,\mathrm{d}x = 0$$
$$\left(\frac{1}{\sqrt{2}}, \sin(j\cdot)\right) = \frac{1}{\sqrt{2}\pi}\int_0^{2\pi} \sin(jx)\,\mathrm{d}x = 0$$

for $j = 1, \ldots, n$, so that the functions in (4.15) form an *orthogonal system*.

The *orthonormality* of the functions in (4.15) finally follows from

$$\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) = \frac{1}{2\pi}\int_0^{2\pi} 1\,\mathrm{d}x = 1$$

and

$$(\cos(j\,\cdot), \cos(j\,\cdot)) = \frac{1}{2\pi}\int_0^{2\pi} [1+\cos(2jx)]\,\mathrm{d}x = 1$$
$$(\sin(j\,\cdot), \sin(j\,\cdot)) = \frac{1}{2\pi}\int_0^{2\pi} [1-\cos(2jx)]\,\mathrm{d}x = 1$$

where we use the representations in (4.16) and (4.17) yet once more. ■

We now connect to the results of Theorems 4.5 and 4.11, where we can, for any function $f \in \mathscr{C}_{2\pi}$, represent its unique best approximation $s^* \in \mathcal{T}_n$ by

$$s^*(x) = \left(f, \frac{1}{\sqrt{2}}\right)\frac{1}{\sqrt{2}} + \sum_{j=1}^{n} \left[(f, \cos(j\cdot))\cos(jx) + (f, \sin(j\cdot))\sin(jx)\right]. \quad (4.19)$$

We reformulate the representation for $s^*$ in (4.19) and introduce on this occasion the important notion of *Fourier partial sums*.

**Corollary 4.12.** *For $f \in \mathscr{C}_{2\pi}$, the unique best approximation $s^* \in \mathcal{T}_n$ to $f$ is given by the $n$-th* `Fourier partial sum` *of $f$,*

$$(F_n f)(x) = \frac{a_0}{2} + \sum_{j=1}^{n} \left[a_j \cos(jx) + b_j \sin(jx)\right]. \quad (4.20)$$

*The coefficients $a_0 = (f, 1)$ and*

$$a_j \equiv a_j(f) = (f, \cos(j\cdot)) = \frac{1}{\pi}\int_0^{2\pi} f(x)\cos(jx)\,\mathrm{d}x \quad (4.21)$$

$$b_j \equiv b_j(f) = (f, \sin(j\cdot)) = \frac{1}{\pi}\int_0^{2\pi} f(x)\sin(jx)\,\mathrm{d}x \quad (4.22)$$

*for $1 \le j \le n$ are called* `Fourier coefficients` *of $f$.* ■

The Fourier partial sum (4.20) is split into an *even* part, given by the partial sum of the *even* trigonometric polynomials $\{\cos(j\cdot), 0 \le j \le n\}$ with "even" Fourier coefficients $a_j$, and into an *odd* part, given by the partial sum of the *odd* trigonometric polynomials $\{\sin(j\cdot), 1 \le j \le n\}$ with "odd" Fourier coefficients $b_j$. We can show that for any even function $f \in \mathscr{C}_{2\pi}$, all odd Fourier coefficients $b_j$ vanish. Likewise, for an odd function $f \in \mathscr{C}_{2\pi}$, all even Fourier coefficients $a_j$ vanish. On this occasion, we recall the result of Proposition 3.42, from which these statements immediately follow. But we wish to compute the Fourier coefficients explicitly.

**Corollary 4.13.** *For $f \in \mathscr{C}_{2\pi}$, the following statements are true.*

*(a) If $f$ is even, then the Fourier partial sum $F_n f$ in (4.20) is even and the Fourier coefficients $a_j$ in (4.21) have the representation*

$$a_j = \frac{2}{\pi} \int_0^{\pi} f(x) \cos(jx) \, \mathrm{d}x \qquad \textit{for } 0 \leq j \leq n.$$

*(b) If $f$ is odd, then the Fourier partial sum $F_n f$ in (4.20) is odd and the Fourier coefficients $b_j$ in (4.22) have the representation*

$$b_j = \frac{2}{\pi} \int_0^{\pi} f(x) \sin(jx) \, \mathrm{d}x \qquad \textit{for } 1 \leq j \leq n.$$

*Proof.* For an even function $f \in \mathscr{C}_{2\pi}$ we have $b_j = 0$, for all $1 \leq j \leq n$, since

$$\begin{aligned} b_j &= \frac{1}{\pi} \int_0^{2\pi} f(x) \sin(jx) \, \mathrm{d}x = -\frac{1}{\pi} \int_0^{-2\pi} f(-x) \sin(-jx) \, \mathrm{d}x \\ &= -\frac{1}{\pi} \int_{-2\pi}^{0} f(x) \sin(jx) \, \mathrm{d}x = -\frac{1}{\pi} \int_0^{2\pi} f(x) \sin(jx) \, \mathrm{d}x = -b_j, \end{aligned}$$

and so the Fourier partial sum $F_n f$ in (4.20) is even. Moreover, we have

$$a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) \, \mathrm{d}x = \frac{2}{\pi} \int_0^{\pi} f(x) \, \mathrm{d}x$$

and, for $1 \leq j \leq n$,

$$\begin{aligned} \pi a_j &= \int_{-\pi}^{\pi} f(x) \cos(jx) \, \mathrm{d}x = \int_{-\pi}^{0} f(x) \cos(jx) \, \mathrm{d}x + \int_0^{\pi} f(x) \cos(jx) \, \mathrm{d}x \\ &= \int_0^{\pi} f(-x) \cos(-jx) \, \mathrm{d}x + \int_0^{\pi} f(x) \cos(jx) \, \mathrm{d}x = 2 \int_0^{\pi} f(x) \cos(jx) \, \mathrm{d}x. \end{aligned}$$

This completes our proof for (a). We can prove (b) analogously. ■

**Example 4.14.** We consider approximating the periodic function $f \in \mathscr{C}_{2\pi}$, defined as $f(x) = \pi - |x|$, for $x \in [-\pi, \pi]$. To this end, we determine for $n \in \mathbb{N}$ the Fourier coefficients $a_j, b_j$ of the Fourier partial sum $F_n f$. Since $f$ is an even function, we can apply Corollary 4.13, statement (a). From this, we see that $b_j = 0$, for all $1 \leq j \leq n$, and, moreover,

$$a_j = \frac{2}{\pi} \int_0^{\pi} f(x) \cos(jx) \, \mathrm{d}x \qquad \text{for } 0 \leq j \leq n.$$

Integration by parts gives

$$\begin{aligned} \int_0^{\pi} f(x) \cos(jx) \, \mathrm{d}x &= \frac{1}{j} f(x) \sin(jx) \Big|_0^{\pi} - \frac{1}{j} \int_0^{\pi} f'(x) \sin(jx) \, \mathrm{d}x \\ &= \frac{1}{j} \int_0^{\pi} \sin(jx) \, \mathrm{d}x = -\frac{1}{j^2} \cos(jx) \Big|_0^{\pi} \qquad \text{for } 1 \leq j \leq n, \end{aligned}$$

and so we have $a_j = 0$ for all *even* indices $j \in \{1, \dots, n\}$, while

$$a_j = \frac{4}{\pi j^2} \qquad \text{for all } \textit{odd} \text{ indices } j \in \{1, \dots, n\}.$$

We finally compute the Fourier coefficient $a_0$ by

$$a_0 = (f, 1) = \frac{1}{\pi} \int_0^{2\pi} f(x)\,\mathrm{d}x = \frac{2}{\pi} \int_0^{\pi} (\pi - x)\,\mathrm{d}x = \frac{2}{\pi} \left[ -\frac{1}{2}(\pi - x)^2 \right]_0^{\pi} = \pi.$$

Altogether, we obtain the representation

$$\begin{aligned} (F_n f)(x) = \frac{\pi}{2} + \sum_{j=1}^{n} a_j \cos(jx) &= \frac{\pi}{2} + \frac{4}{\pi} \sum_{\substack{j=1 \\ j \text{ odd}}}^{n} \frac{1}{j^2} \cos(jx) \\ &= \frac{\pi}{2} + \frac{4}{\pi} \sum_{k=0}^{\lfloor \frac{n-1}{2} \rfloor} \frac{\cos((2k+1)x)}{(2k+1)^2} \end{aligned}$$

for the $n$-th Fourier partial sum of $f$.

For illustration the function graphs of the Fourier partial sums $F_n f$ and of the error functions $F_n f - f$, for $n = 2, 4, 16$, are shown in Figures 4.1-4.3. ♢

As we have seen in Section in 2.6, the *real-valued* Fourier partial sum $F_n f$ in (4.20) can be represented as *complex* Fourier partial sum of the form

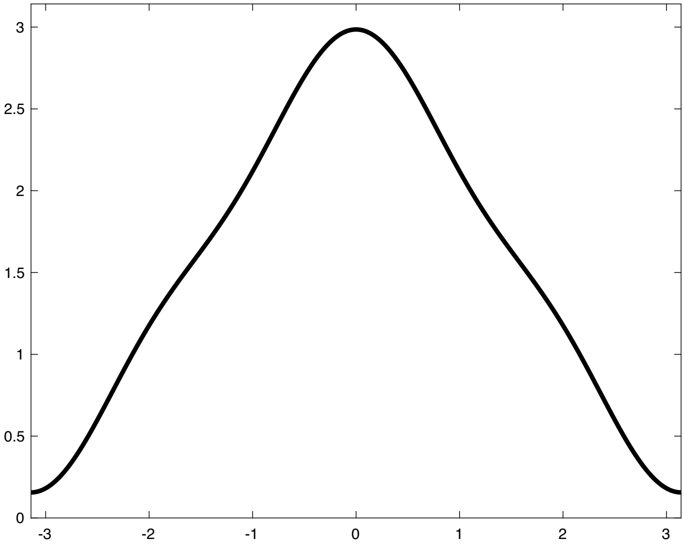
$$(F_n f)(x) = \sum_{j=-n}^{n} c_j \mathrm{e}^{\mathrm{i}jx}. \tag{4.23}$$

For the conversion of the Fourier coefficients, we apply the linear mapping in (2.69), whereby, with using the Eulerean formula (2.67), we obtain for the complex Fourier coefficients in (4.23) the representation

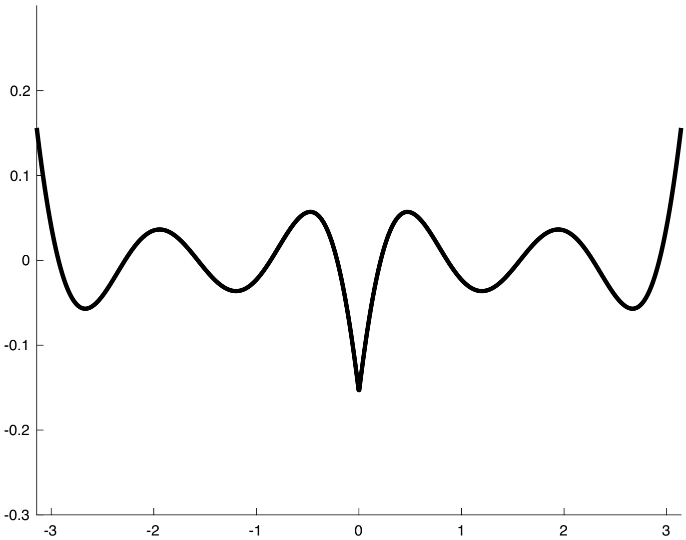
$$c_j = \frac{1}{2\pi} \int_0^{2\pi} f(x) \mathrm{e}^{-\mathrm{i}jx}\,\mathrm{d}x \qquad \text{for } j = -n, \dots, n. \tag{4.24}$$

We remark that the complex Fourier coefficients $c_j$ in (4.24) can also, like the real Fourier coefficients $a_j$ in (4.21) and $b_j$ in (4.22), be expressed via inner products. In fact, by using the *complex* inner product $(\cdot, \cdot)_{\mathbb{C}}$ in (4.14), we can rewrite the representation in (4.24) as

$$c_j = (f, \exp(\mathrm{i}j\cdot))_{\mathbb{C}} \qquad \text{for } j = -n, \dots, n.$$

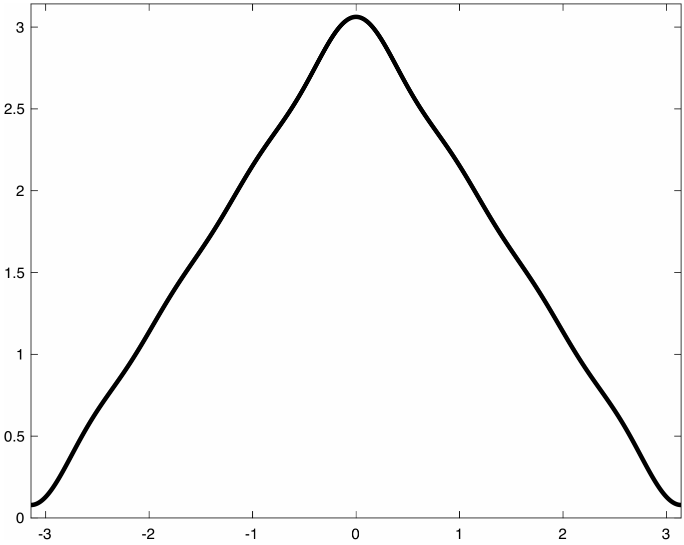



Fourier partial sum $F_2 f$

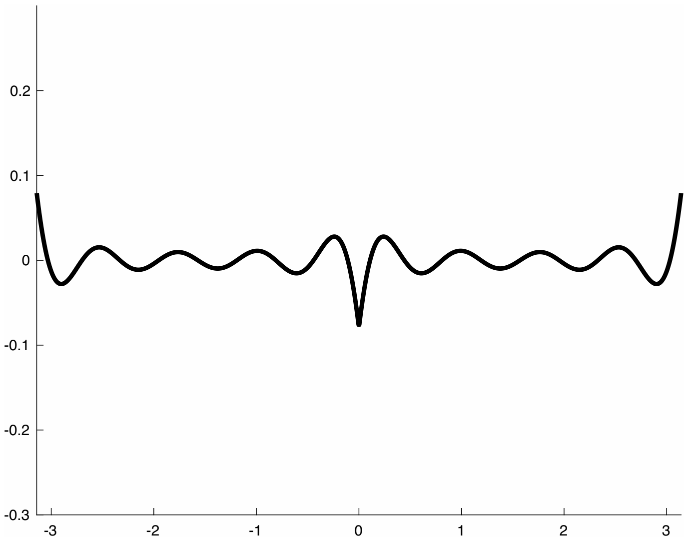



error function $F_2 f - f$

**Fig. 4.1.** Approximation of the function $f(x) = \pi - |x|$ on the interval $[-\pi, \pi]$ by the Fourier partial sum $(F_2 f)(x)$ (see Example 4.14).

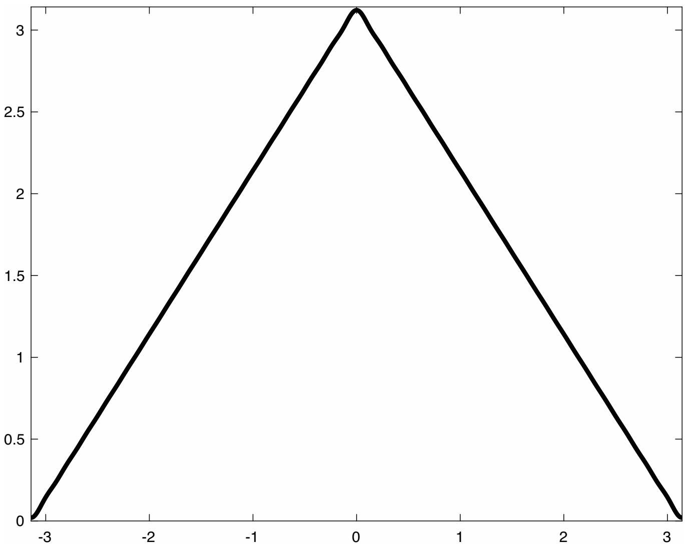



Fourier partial sum $F_4 f$

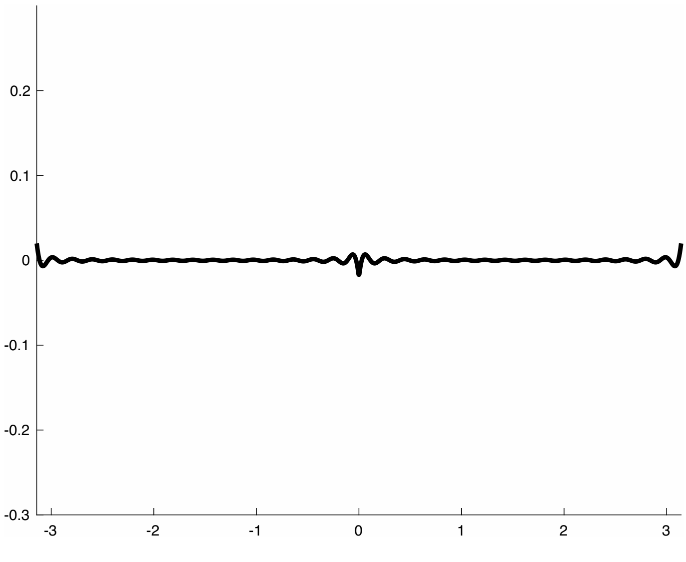



error function $F_4 f - f$

**Fig. 4.2.** Approximation of the function $f(x) = \pi - |x|$ on the interval $[-\pi, \pi]$ by the Fourier partial sum $(F_4 f)(x)$ (see Example 4.14).




Fourier partial sum $F_{16}f$




error function $F_{16}f - f$

**Fig. 4.3.** Approximation of the function $f(x) = \pi - |x|$ on the interval $[-\pi, \pi]$ by the Fourier partial sum $(F_{16}f)(x)$ (see Example 4.14).

Now we wish to approximate the complex Fourier coefficients $c_j$.

To this end, we apply the *composite trapezoidal rule* with $N = 2n + 1$ equidistant knots

$$x_k = \frac{2\pi}{N} k \in [0, 2\pi) \qquad \text{for } k = 0, \ldots, N-1,$$

so that

$$c_j \approx \frac{1}{N} \sum_{k=0}^{N-1} f(x_k) \mathrm{e}^{-\mathrm{i} j x_k} = \frac{1}{N} \sum_{k=0}^{N-1} f(x_k) \omega_N^{-jk} \tag{4.25}$$

where $\omega_N = \mathrm{e}^{2\pi\mathrm{i}/N}$ denotes the $N$-th root of unity in (2.74). In this way, the vector $c = (c_{-n}, \ldots, c_n)^T \in \mathbb{C}^N$ of the complex Fourier coefficients (4.24) is approximated by the discrete Fourier transform (2.79) from the data vector

$$f = (f_0, \ldots, f_{N-1})^T \in \mathbb{R}^N,$$

where $f_k = f(x_k)$ for $k = 0, \ldots, N-1$. In order to compute the Fourier coefficients $c \in \mathbb{C}^N$ efficiently, we apply the fast Fourier transform (FFT) from Section 2.7. According to Theorem 2.46, the FFT can be performed in $\mathcal{O}(N \log(N))$ steps.

We close this section with the following remark.

The `Fourier operator` $F_n : \mathscr{C}_{2\pi} \longrightarrow \mathcal{T}_n$ gives the orthogonal projection of $\mathscr{C}_{2\pi}$ onto $\mathcal{T}_n$. In Chapter 6, we will analyze the asymptotic behaviour of the operator $F_n$, for $n \to \infty$, in more detail, where we will address the following fundamental questions.

- Is the `Fourier series`

$$(F_\infty f)(x) = \frac{a_0}{2} + \sum_{j=1}^{\infty} [a_j \cos(jx) + b_j \sin(jx)] \qquad \text{for } f \in \mathscr{C}_{2\pi}$$

  of $f$ convergent?
- If so, does the Fourier series $F_\infty f$ converge to $f$?
- If so, how fast does the Fourier series $F_\infty f$ converge to $f$?

In particular, we will investigate, if at all, *in which sense* (e.g. pointwise, or uniformly, or with respect to the Euclidean norm $\| \cdot \|$) the convergence of the Fourier series $F_\infty f$ holds. In Chapter 6, we will give answers, especially for the asymptotic behaviour of the approximation error

$$\eta(f, \mathcal{T}_n) = \|F_n f - f\| \quad \text{and} \quad \eta_\infty(f, \mathcal{T}_n) = \|F_n f - f\|_\infty \qquad \text{for } n \to \infty.$$

This will lead us to specific conditions on the smoothness of $f$.